

Data Mining Chapter 4 Classification

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Question 5

We now examine the differences between LDA and QDA.

- (a) If the Bayes decision boundary is linear, do we expect LDA or QDA to perform better on the training set? On the test set?

Answer

On the training data: We would expect QDA to perform better, as it has more variance than LDA. Therefore, it will fit the training data better, resulting in a lower training error.

On the test data: We would expect LDA to perform better than QDA, with a lower test error, since the true Bayes decision boundary is linear.

- (b) If the Bayes decision boundary is non-linear, do we expect LDA or QDA to perform better on the training set? On the test set?

Answer

On the training data: We would expect QDA to perform better, as it has more variance than LDA. Therefore, it will fit the training data better, resulting in a lower training error.

On the test data: We would expect QDA to perform better than LDA, with a lower test error, since the true Bayes decision boundary is non-linear. QDA is more flexible and is therefore better able to capture the non-linearity.

- (c) In general, as the sample size n increases, do we expect the test prediction accuracy of QDA relative to LDA to improve, decline, or be unchanged? Why?

Answer

As the sample size n increases, we expect the test prediction accuracy of QDA relative to LDA to **improve**. QDA has higher variance than LDA because it estimates a separate covariance matrix for each class. With a larger sample size, these covariance matrices can be estimated more accurately, reducing the variance of QDA. Therefore, QDA can take advantage of its greater flexibility and potentially provide better predictions.

- (d) True or False: Even if the Bayes decision boundary for a given problem is linear, we will probably achieve a superior test error rate using QDA rather than LDA

because QDA is flexible enough to model a linear decision boundary. Justify your answer.

Answer

False.

Even though QDA is flexible enough to represent a linear decision boundary, if the true Bayes decision boundary is linear, LDA will generally perform better on the test data. This is because LDA has lower variance since it assumes that all classes share the same covariance matrix.

QDA estimates a separate covariance matrix for each class, which introduces additional parameters and therefore higher variance. If the true boundary is linear, this additional flexibility is unnecessary and may lead to overfitting. Thus, LDA is generally expected to achieve a lower test error in this situation.

Question 10

Equation 4.32 derived an expression for

$$\log \left(\frac{\Pr(Y = k \mid X = x)}{\Pr(Y = K \mid X = x)} \right)$$

in the setting where $p > 1$, so that the mean for the k th class, μ_k , is a p -dimensional vector, and the shared covariance Σ is a $p \times p$ matrix.

However, in the setting with $p = 1$, (4.32) takes a simpler form, since the means $\mu_1, \dots, \mu_K$ and the variance σ^2 are scalars.

In this simpler setting, repeat the calculation in (4.32), and provide expressions for a_k and b_{kj} in terms of π_k , π_K , μ_k , μ_K , and σ^2 .

Answer

For the case where $p = 1$, we derive an expression for

$$\log \left(\frac{\Pr(Y = k \mid X = x)}{\Pr(Y = K \mid X = x)} \right).$$

Since LDA assumes that all classes share the same variance σ^2 , Bayes' theorem gives

$$\log \left(\frac{\Pr(Y = k \mid X = x)}{\Pr(Y = K \mid X = x)} \right) = \log \left(\frac{\pi_k f_k(x)}{\pi_K f_K(x)} \right),$$

where π_k and π_K are the prior probabilities for classes k and K , respectively, and $f_k(x)$ and $f_K(x)$ are Gaussian density functions.

For the univariate Gaussian distribution,

$$f_k(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{1}{2\sigma^2}(x - \mu_k)^2 \right].$$

Therefore,

$$\log \left(\frac{\Pr(Y = k \mid X = x)}{\Pr(Y = K \mid X = x)} \right) = \log \left(\frac{\pi_k e^{-\frac{1}{2\sigma^2}(x-\mu_k)^2}}{\pi_K e^{-\frac{1}{2\sigma^2}(x-\mu_K)^2}} \right).$$

Taking the logarithm gives

$$= \log \left(\frac{\pi_k}{\pi_K} \right) + \frac{1}{2\sigma^2} [-2x\mu_K + \mu_K^2 + 2x\mu_k - \mu_k^2].$$

Collecting the constant terms and the terms involving x ,

$$= \log \left(\frac{\pi_k}{\pi_K} \right) + \frac{\mu_K^2 - \mu_k^2}{2\sigma^2} + \frac{\mu_k - \mu_K}{\sigma^2} x.$$

This expression is linear in x and can therefore be written as

$$\log \left(\frac{\Pr(Y = k \mid X = x)}{\Pr(Y = K \mid X = x)} \right) = a_k + b_k x,$$

where

$$a_k = \log \left(\frac{\pi_k}{\pi_K} \right) + \frac{\mu_K^2 - \mu_k^2}{2\sigma^2}$$

and

$$b_k = \frac{\mu_k - \mu_K}{\sigma^2}.$$

Thus, when $p = 1$, the log-odds between class k and the reference class K are a linear function of x . This is consistent with the linear decision boundary assumed by Linear Discriminant Analysis.

Question 14

In this problem, you will develop a model to predict whether a given car gets high or low gas mileage based on the `Auto` data set.

- (a) Create a binary variable, `mpg01`, that contains a 1 if `mpg` contains a value above its median, and a 0 if `mpg` contains a value below its median.

You can compute the median using the `median()` method of the data frame. You may find it helpful to add a column `mpg01` to the data frame by assignment. Assuming you have stored the data frame as `Auto`, this can be done as follows:

```
Auto['mpg01'] = mpg01
```

Answer

A binary response variable, `mpg01`, was created based on the median value of `mpg`. It was defined as

$$\text{mpg01} = \begin{cases} 1, & \text{if } \text{mpg} > \text{median}(\text{mpg}), \\ 0, & \text{otherwise.} \end{cases}$$

Thus, cars with gas mileage above the median were classified as high mileage cars, while the remaining cars were classified as low mileage cars.

- (b) Explore the data graphically in order to investigate the association between `mpg01` and the other features.

Answer

Boxplots and the correlation matrix were used to investigate the association between `mpg01` and the available predictors.

The variables that appeared to be most useful for predicting `mpg01` were:

- Cylinders,
- Displacement,
- Horsepower,
- Weight.

The correlations between these variables and `mpg01` were approximately

$$\begin{aligned} \text{Cor}(\text{cylinders}, \text{mpg01}) &= -0.759, \\ \text{Cor}(\text{displacement}, \text{mpg01}) &= -0.753, \\ \text{Cor}(\text{horsepower}, \text{mpg01}) &= -0.667, \\ \text{Cor}(\text{weight}, \text{mpg01}) &= -0.758. \end{aligned}$$

These negative correlations indicate that cars with larger engines, more cylinders, greater horsepower, and greater weight tend to have lower gas mileage.

It is also important to note that these predictors are strongly correlated with each other. For example, the correlation between cylinders and displacement is approximately 0.951, while that between displacement and weight is approximately 0.933.

Which of the other features seem most likely to be useful in predicting `mpg01`?

Scatterplots and boxplots may be useful tools to answer this question. Describe your findings.

- (c) Split the data into a training set and a test set.

Answer

The observations were randomly divided into training and test sets. The test set contained 25% of the observations, while the remaining 75% were used for training.

For the subsequent classification models, `displacement` and `weight` were selected as predictors.

- (d) Perform LDA on the training data in order to predict `mpg01` using the variables that seemed most associated with `mpg01` in (b). What is the test error of the model obtained?

Answer

Linear Discriminant Analysis (LDA) was fitted using `displacement` and `weight`.

The resulting confusion matrix was

	True 0	True 1
Predicted 0	37	3
Predicted 1	11	47

Therefore, the classification accuracy was

$$\frac{37 + 47}{98} = 0.8571 \approx 85.7\%.$$

The corresponding test error was

$$1 - 0.8571 = 0.1429,$$

or approximately

$$\boxed{14.3\%}.$$

- (e) Perform QDA on the training data in order to predict `mpg01` using the variables that seemed most associated with `mpg01` in (b). What is the test error of the model obtained?

Answer

QDA was also fitted using `displacement` and `weight`. The confusion matrix obtained was

	True 0	True 1
Predicted 0	39	7
Predicted 1	9	43

The classification accuracy was

$$\frac{39 + 43}{98} = 0.8367 \approx 83.7\%.$$

Hence, the test error was

$$1 - 0.8367 = 0.1633,$$

or approximately

$$\boxed{16.3\%}.$$

- (f) Perform logistic regression on the training data in order to predict `mpg01` using the variables that seemed most associated with `mpg01` in (b). What is the test error of the model obtained?

Answer

A logistic regression model was fitted using `displacement` and `weight`. Predicted probabilities greater than 0.5 were classified as class 1.

The confusion matrix was

	True 0	True 1
Predicted 0	38	7
Predicted 1	10	43

The classification accuracy was

$$\frac{38 + 43}{98} = 0.8265 \approx 82.7\%.$$

Therefore, the test error was

$$1 - 0.8265 = 0.1735,$$

or approximately

$$\boxed{17.3\%}.$$

- (g) Perform naive Bayes on the training data in order to predict `mpg01` using the variables that seemed most associated with `mpg01` in (b). What is the test error of the model obtained?

Answer

A Gaussian Naive Bayes classifier was fitted using the same two predictors.

The confusion matrix obtained was

	True 0	True 1
Predicted 0	38	6
Predicted 1	10	44

The resulting classification accuracy was

$$\frac{38 + 44}{98} = 0.8367 \approx 83.7\%.$$

Thus, the test error was approximately

$$\boxed{16.3\%}.$$

- (h) Perform KNN on the training data, with several values of K , in order to predict `mpg01`. Use only the variables that seemed most associated with `mpg01` in (b). What test errors do you obtain? Which value of K seems to perform the best on this data set?

Answer

KNN was fitted using `displacement` and `weight` for several values of K .

Among the values examined, $K = 2$ produced the best performance. Its confusion matrix was

	True 0	True 1
Predicted 0	44	9
Predicted 1	4	41

The accuracy was

$$\frac{44 + 41}{98} = 0.8673 \approx 86.7\%.$$

Therefore, the test error was

$$1 - 0.8673 = 0.1327,$$

or approximately

$$\boxed{13.3\%}.$$

Hence, among the KNN models examined,

$$\boxed{K = 2}$$

gave the best performance.

Overall Comparison

The classification results can be summarized as follows:

Model	Accuracy	Test Error
LDA	85.7%	14.3%
QDA	83.7%	16.3%
Logistic Regression	82.7%	17.3%
Naive Bayes	83.7%	16.3%
KNN ($K = 2$)	86.7%	13.3%

Therefore, based on the test results obtained in this analysis, KNN with $K = 2$ achieved the highest classification accuracy and the lowest test error.

Question 16

Using the `Boston` data set, fit classification models in order to predict whether a given suburb has a crime rate above or below the median.

Explore logistic regression, LDA, naive Bayes, and KNN models using various subsets of the predictors.

Describe your findings.

Hint: You will have to create the response variable yourself, using the variables that are contained in the `Boston` data set.

Answer

Creation of the Response Variable

A binary response variable called `crim01` was created from the `crim` variable:

$$\text{crim01} = \begin{cases} 1, & \text{if } \text{crim} > \text{median}(\text{crim}), \\ 0, & \text{otherwise.} \end{cases}$$

Therefore, class 1 represents suburbs with crime rates above the median, while class 0 represents suburbs with crime rates at or below the median.

Exploratory Analysis

Boxplots and correlations were examined to determine which predictors were most strongly associated with `crim01`.

The variables that appeared most useful were

- `nox`,
- `age`,
- `indus`,
- `dis`,
- `rad`,
- `tax`.

Their correlations with `crim01` were approximately

$$\begin{aligned} \text{Cor}(\text{nox}, \text{crim01}) &= 0.723, \\ \text{Cor}(\text{age}, \text{crim01}) &= 0.614, \\ \text{Cor}(\text{indus}, \text{crim01}) &= 0.603, \\ \text{Cor}(\text{dis}, \text{crim01}) &= -0.616, \\ \text{Cor}(\text{rad}, \text{crim01}) &= 0.620, \\ \text{Cor}(\text{tax}, \text{crim01}) &= 0.609. \end{aligned}$$

Among these variables, `nox` showed the strongest association with the binary crime response.

Training and Test Sets

The data were divided into training and test sets, with 25% of the observations assigned to the test set.

Logistic regression, LDA, QDA, Naive Bayes, and KNN were then compared using different subsets of predictors.

Models Using All Predictors

When all available predictors were used, the following test accuracies were obtained:

Model	Accuracy
Logistic Regression	85.04%
LDA	82.68%
QDA	91.34%
Naive Bayes	77.95%
KNN ($K = 1$)	92.91%
KNN ($K = 2$)	91.34%
KNN ($K = 5$)	89.76%
KNN ($K = 10$)	85.83%
KNN ($K = 15$)	84.25%
KNN ($K = 20$)	81.89%
KNN ($K = 25$)	79.53%

The best result using all predictors was obtained by KNN with $K = 1$, which achieved an accuracy of

92.91%.

Models Using `nox`, `age`, `indus`, `dis`, `rad`, and `tax`

Using the six predictors that appeared most strongly associated with `crim01`, the following results were obtained:

Model	Accuracy
Logistic Regression	86.61%
LDA	81.10%
QDA	90.55%
Naive Bayes	79.53%
KNN ($K = 1$)	90.55%
KNN ($K = 2$)	89.76%
KNN ($K = 5$)	91.34%
KNN ($K = 10$)	85.04%
KNN ($K = 15$)	81.89%
KNN ($K = 20$)	80.31%
KNN ($K = 25$)	79.53%

For this predictor subset, KNN with $K = 5$ performed best, achieving

$$\boxed{91.34\%}$$

accuracy.

Models Using nox and rad

Using only nox and rad produced particularly strong KNN results:

Model	Accuracy
Logistic Regression	82.68%
LDA	81.89%
QDA	82.68%
Naive Bayes	84.25%
KNN ($K = 1$)	94.49%
KNN ($K = 2$)	93.70%
KNN ($K = 5$)	93.70%
KNN ($K = 10$)	92.91%
KNN ($K = 15$)	92.91%
KNN ($K = 20$)	92.91%
KNN ($K = 25$)	92.13%

The best-performing model in this comparison was KNN with $K = 1$:

$$\boxed{\text{Accuracy} = 94.49\%}.$$

Its corresponding test error was approximately

$$100\% - 94.49\% = \boxed{5.51\%}.$$

Additional Predictor Subsets

Additional combinations of predictors were also investigated.

Using nox and ptratio, KNN with $K = 1$ produced the highest accuracy:

$$\boxed{92.13\%}.$$

Using rad and ptratio, KNN with $K = 1$, $K = 2$, and $K = 5$ each achieved an accuracy of approximately

$$\boxed{93.70\%}.$$

Finally, when only nox was used, KNN with $K = 2$ and $K = 5$ both achieved approximately

$$\boxed{93.70\%}$$

accuracy.

Conclusion

The results show that the choice of predictors and classification method has a substantial effect on prediction accuracy.

Among all the models and predictor subsets examined, the best performance was obtained using

KNN with $K = 1$

with the predictors

`nox` and `rad`.

This model achieved a test accuracy of approximately

94.49%

and a corresponding test error of approximately

5.51%.

Therefore, the analysis suggests that `nox` and `rad` contain substantial information for distinguishing suburbs with high and low crime rates, and KNN was the strongest classifier among the models investigated in the notebook.